

D-CORE: Diagnosing and Repairing Distributed Information Failures in Long-Horizon Agent Teams

Anonymous Author(s)

Submission Id: Draft; not submitted

Abstract

A distributed agent can take an inappropriate action because it ignores available evidence, a teammate never communicates an observation, or a correct warning arrives too late. An action sequence and final score need not distinguish these failures or show whether a legal intervention could change the outcome. We present D-CORE, a specification-based method that reconstructs decision-time evidence, emits structured diagnostic witnesses, and maps supported failures to bounded repairs. Append-only execution evidence and deterministic prefix reconstruction establish a common checkpoint; unchanged continuations test replay equivalence, while repaired continuations discard cached futures and resume under the original remaining budget. We prescribe three full-season soybean scenarios, capability-typed diagnostic comparators, 120 independently annotated decisions, 60 repair checkpoints, and five selective live-verification policies. Outcomes retain missingness and distinguish harvest, transfer, storage, and postharvest compliance. Empirical estimates and comparative conclusions remain pending the frozen study.

Keywords

Multiagent systems, agent debugging, failure diagnosis, repair, deterministic replay

ACM Reference Format:

Anonymous Author(s). 2027. D-CORE: Diagnosing and Repairing Distributed Information Failures in Long-Horizon Agent Teams. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 7 pages.

1 Introduction

Distributed tool-using agents divide observation, interpretation and execution across roles. This division introduces an evaluation problem: the correctness of a decision depends both on the state of the world and on the evidence available to its owner. A sensing agent may detect disease while an operations agent retains an earlier assessment. A warning can be omitted, delayed, delivered outside its validity period, or applied to the wrong region. These are different failures even when they produce the same attempted treatment and final outcome.

Long-horizon environments make the distinction consequential. A native tool may reject an action immediately, while the missed

opportunity appears only at harvest. Alternatively, a communication failure may have no measurable physical cost because the action remains inside an execution window or a later intervention recovers the loss. An evaluator should retain the original failure and its recovery without treating either as proof of a yield effect. It should also distinguish an invalid proposal from failed native execution and from deliberate abstention.

D-CORE evaluates distributed information failures under an explicit specification. Its central objects are an execution partial order, a normative occurrence net, versioned local evidence and authoritative simulator outcomes. The evaluator reconstructs its judgments from saved evidence rather than accepting the runtime guard’s verdict as ground truth. It reports event fidelity, semantic obligation conformance, local and global disagreement and the earliest identifiable break in an information path. A weighted scalar is secondary to this decomposition.

The intended contributions are fourfold. First, we give an operational contract connecting role-specific tools, temporally eligible evidence and exact native receipts to a concurrency-aware specification. Second, we produce prerequisite-level diagnostic witnesses that distinguish missing observation, delivery, freshness, scope, context, reasoning and execution failures while preserving unresolved cases. Third, we select at most two legal, timely repair primitives and test them through matched native continuations from a verified prefix. Fourth, we evaluate selective live verification under the same prefix evidence and team budget as audit and fixed-policy baselines. Whether diagnosis or repair helps is empirical; the study retains ties, failures and adverse effects.

The farm environment is reused. D-CORE does not introduce a new crop simulator, establish agronomic treatment efficacy or claim that a provenance path identifies physical causation. Its proposed advance is the ability to assess what different team members could support at a decision and to locate information failures that a projected action log cannot identify.

2 Related Work and Relationship to Prior Systems

Full-path agent evaluation. CORE encodes permissible tool-use paths using deterministic finite automata and evaluates behavior beyond final-state success [7]. D-CORE retains full-path evaluation as a foundation. Its additional objects describe concurrent roles, distributed evidence eligibility and information-path localization. Sequential path metrics remain comparison outputs; changing the notation from an automaton to a Petri net is not itself the contribution.

FarmARE and FAIRY. The earlier farm evaluation study examines full-season soybean operations and nine controller methodologies [8]. The FAIRY system paper describes the event-driven



This work is licensed under a Creative Commons Attribution International 4.0 License.

farm engine, sensing and machinery interfaces, controller backends, scenario suite and spatiotemporal correctness evaluation [9]. Its public preprint reports acceptance to ACM SIGSPATIAL 2026; we cite the available preprint without inventing proceedings identifiers. D-CORE reuses the native scenarios, physics, tools and controller implementations, and distinguishes assigned A2A routing from observed delegation. Its new experimental variables concern distributed information and communication faults. Historical farm results are context, not new paired observations. We do not attribute the earlier system’s deployment or physical calibration evidence to D-CORE.

Multiagent benchmarks and taxonomies. MultiAgentBench assesses collaboration and competition through tasks and milestone indicators [13]. MAST studies multiagent failures through annotated traces and a failure taxonomy [2]. These establish that coordination assessment and human failure labels are existing research directions. D-CORE asks a narrower question: can an executable specification and recorded fact-version paths support reproducible, information-relative diagnoses? Independent annotation is necessary to test this operational claim; a self-consistent evaluator is insufficient evidence of diagnostic validity.

Failure attribution and intervention. Who&When formulates agent/step attribution and shows that exact failure-step localization remains difficult [12]. AgentRx derives and checks trajectory constraints before an LLM judge localizes a failure [1]. DCFA combines global dependency construction with local counterfactual-style reasoning [10]; because its repository exposes no license in our frozen source review, our clean implementation is labeled a DCFA-style reimplement. DoVer evaluates debugging hypotheses through interventions rather than log-only attribution [6]. D-CORE adopts the need for intervention-based evidence but constrains each intervention by actor ownership, prefix availability, scope, time, native cost and the original budget. MARBLE’s agricultural milestones provide a path-level comparator, not a failure diagnosis or repair method [13].

Concurrency and conformance. Happens-before provides a partial order for distributed execution [4]; event structures relate enabling and consistency to concurrent processes [11]. Conformance checking with partially ordered event data already addresses limitations of total-order logs [5]. D-CORE builds on these foundations and uses a restricted hierarchical workflow compiled to a labeled, 1-safe occurrence net. It does not implement arbitrary timed or data Petri nets. Finally, identifying an actual cause requires a counterfactual model and assumptions beyond trace reachability [3]. Our localization labels concern recorded provenance. Paired simulator interventions answer a separate question about outcome effects.

3 Setting and Motivating Example

Let A be a fixed team of actors, each with an exclusive set of mutable tool grants and a set of observable tools. The environment advances in simulated world time. Agents receive only their local observations, their own action results and messages delivered along permitted routes. A shared-blackboard condition changes information availability while retaining the aggregate tool set. Capability cards expose teammate responsibilities and legal communication

routes. A refusal states why a requested action is unavailable; another role may perform that exact requested action after a valid handoff. The runtime never substitutes an unrelated permissible action.

Consider a hypothetical spray decision at time 4. At time 0, an operator receives a forecast supporting spraying through time 5. Actual weather changes at time 1, and a sensing agent observes the closure at time 2. In execution A, the closure warning reaches the operator before time 4; in execution B, a recorded send exists but timely delivery does not. Hold the proposed spray and native result fixed. Under a specification in which a current warning overrides a forecast, A contradicts available evidence, whereas B exposes a communication-path failure. These times illustrate order, not calibrated agronomic thresholds. No yield difference is implied.

The same-evidence flat baseline can distinguish available facts in this example. The example therefore motivates retaining evidence, not superiority over that baseline. Conversely, an action-only projection loses the distinguishing send, receive and snapshot records. If the send itself is unavailable, the evaluator cannot decide whether the warning was omitted or lost in transit. It must retain an unresolved diagnosis.

4 Formal Objects and Evaluation

4.1 Trace, facts and local information

An event is a tuple containing a stable identifier, owner, kind, logical and world times, tool and arguments where applicable, scope, status and recorded parents. Let $T = (E, <)$ be the trace, with $<$ generated by actor order, matching send/receive edges and recorded action-effect dependencies. Vector timestamps help audit this relation; they do not establish physical causation.

A fact version is

$$f = (k, v, u, s, o, t_o, t_l, t_x, v, \pi),$$

where k is a key, v a value, u its units, s its scope, o the originating observation, t_o observation time, t_l learned time, t_x expiry, v version information and π supporting identifiers. Units are declared in the fact registry and preserved with values. Scope is a field, an inclusive ridge interval or a named resource/equipment scope. Sensor identifiers are opaque: a sensor named S3 does not imply ridge 3. Returned coverage determines which ridges its evidence supports.

The immutable pre-decision snapshot $K_a(d)$ contains exactly the evidence made available to actor a before decision d . A fact is eligible only if the observation and delivery precede the decision, its scope satisfies the requirement, its validity has not expired, and no applicable version supersedes it. Observation time governs freshness; a reordered old message does not become fresh because it arrived last. Supersession may be regional, so newer evidence for one interval need not erase evidence for disjoint ridges. Missing, unsupported or expired evidence produces *unknown*, not a fabricated false observation.

Authoritative world versions are recorded separately for evaluation and are excluded from actor prompts. Receipt means availability, not cognitive uptake. The evaluator cannot infer an agent’s internal beliefs from its text, and an absent snapshot item supports only a recorded uptake diagnosis.

For long-horizon execution, the distributed controller retains bounded actor-local memory of accepted planting, replanting and harvest requests across prompt pruning. Entries preserve arguments, scope, receipt identifiers and result times; they do not certify current crop state or reveal teammate actions. Remaining request allocation is visible to the controller. This memory aid is part of the declared adapter, and its presence does not establish successful planning or a physical benefit.

4.2 Specification and nonanticipating policy

A specification S declares actors, phase windows, transition acceptance, branches, information policies, negative obligations, semantic causal obligations and fault treatments. Each scientific choice is traced to a native mechanic, published support or an explicit modeling assumption. Source code is evidence for simulator behavior, not independent agronomic validation.

A local information policy is a total, disjoint map

$$\Pi_{a,p} : \{T, F, U\}^q \times \{\text{open}, \text{closed}\}^2 \longrightarrow (\mathcal{P}, \mathcal{R}),$$

where p denotes a phase, q its declared requirements, and the two binary states represent deadline and channel state. $\mathcal{P}$ and $\mathcal{R} \subseteq \mathcal{P}$ are permitted and required response classes: execute, reobserve, handoff, defer or abstain. Validation enumerates the finite input space and rejects missing or overlapping rules. The evaluator reconstructs the input vector from the decision snapshot and uses world time for the deadline. Runtime policy commitments are redundant audit records, not evaluator labels.

World branches use only authoritative versions eligible at their declared commitment phase. They include an explicit complement and cannot depend on a later action or outcome. Missing or ambiguous branch evidence invalidates a definitive branch judgment. This prevents an evaluator from selecting whichever normative workflow best fits an observed successful trace.

Specifications follow either independent review with adjudication or the declared author-defined/professor-approved route. Author-written specifications are never labeled independently confirmed. Professor approval binds exact process, team, role-refinement and protocol digests. Any scientific revision requires renewed approval. Independent episode annotation answers a different question and remains independent under both specification routes.

The authored farm contracts distinguish a regional aggregate from evidence about a subregion: positive disease and mean-water predicates require the declared region, while an upper bound on the maximum moisture can cover a smaller harvest batch. Surface moisture is an explicitly imperfect information proxy for evaluator-only root stress; hidden root measurements never enter an agent’s local state. The primary maximum evidence age is 24 hours, with a 12-hour alternative frozen before confirmation. Both also respect observation-time expiry. Calendar windows are author-defined protocol choices, and cultivar-specific windows permit harvest and storage to interleave. These choices require scientific review rather than deriving their validity from native task completion.

4.3 Action acceptance and event fidelity

Native execution is authoritative for whether a tool acted. A request-bound receipt records the exact actor, intent, tool, arguments, native event identifier, success/error and result digest. A failed action does not satisfy a successful transition. An accepted retry is a new observed attempt linked to the earlier failure. A blocked proposal has no native write identifier, and tests verify nonmutation. An uncertain interrupted write is preserved for inspection rather than blindly replayed.

For each required transition t , categorical acceptance predicates cover owner/tool, critical arguments, spatial overlap, time window and successful status. Let c_t indicate that all critical predicates pass under the evaluator’s matching. Within module m , the positive coverage contribution is the weighted fraction of accepted required transitions. Frozen negative obligations supply penalties for prohibited or unnecessary unmatched actions. Module normalization prevents an expanded ridge batch from acquiring extra importance merely because it contains more native calls. The event-fidelity profile reports overall fidelity together with presence, argument, scope, timing and execution acceptance. Missing transitions reduce fidelity; unavailable final yield remains unavailable.

Exact-match argument constraints may be too restrictive for legitimate adaptive workflows. Accordingly, tolerances and acceptable alternatives must be declared before evaluation and examined in specification sensitivity. The evaluator does not silently repair arguments to an oracle value. Unmatched behavior that is neither accepted nor covered by a negative classifier remains unresolved.

4.4 Semantic obligation conformance

An obligation o has a target, weight and predeclared alternative realizations. Each realization contains required order edges, data guards and, where needed, an actor route. Let $z_o = 1$ if at least one complete realization holds. Causal conformance (CC) is a module-normalized weighted average of applicable z_o values. The name denotes normative dependency conformance; it is not an estimate of an actual cause.

One end-to-end obligation remains one denominator item when a communication path is refined into additional hops. Hop-level records can improve localization without increasing the weight of the scientific obligation. Missing target actions are charged by event fidelity; CC reports the applicable-target denominator, and a module with no applicable obligation is NA. Reporting both profiles prevents high conditional CC on a tiny completed prefix from masquerading as full-season success. If a branch lacks phase-specific authoritative evidence, its selection is unknown: only obligations shared by all alternatives are assessable, with branch-dependent exclusions reported explicitly. Missing common obligations still count against event fidelity.

The secondary score is $D = \alpha EF + (1 - \alpha) CC$, with $\alpha = 0.5$ primary and prespecified 0.25/0.75 sensitivity. The weights are not selected to maximize association with yield. Diagnostic decomposition, coverage and uncertainty accompany every scalar.

4.5 Local/global disagreement and localization

For an assessable high-impact decision, let $L(d)$ indicate that its response conforms to the reconstructed local policy and $G(d)$ indicate global action conformance against authoritative facts and the specification. The evaluator reports the weighted two-by-two table and information/global discordance

$$IGD = \frac{\sum_{d \in \mathcal{D}} w_d \mathbf{1}[L(d) = 1, G(d) = 0]}{\sum_{d \in \mathcal{D}} w_d}.$$

$\mathcal{D}$ contains decisions with the evidence needed for the comparison. Unknown local and global cases and decisions outside policy coverage are counted separately; they are not recoded as agreement.

Localization follows identifiers along the chain from authoritative version to observation, claim, delivery, forwarded version, decision snapshot and native action. Each failed prerequisite yields a witness containing the decision, obligation, actor, scope, validity rule, supporting events and determination status. Witnesses distinguish missing observation, failed delivery, expired evidence, incorrect scope, received evidence omitted from context, failure to use available evidence and native execution failure. Labels can overlap and share a root-support group. Missing or unverifiable support remains unresolved rather than being forced into a known mechanism.

A recovery retains the original rejection, later evidence and subsequent actions. Guard effects are evaluated only for assessable proposals. A physical prevention requires enforcement, a non-allow verdict and evidence that all linked attempts were blocked before native execution. Counting every audit verdict as a prevented write would manufacture a treatment benefit.

4.6 Prefix reconstruction and bounded repair

A durable journal flushes each model request and response, parsed or rejected proposal, native-write intent and receipt, delivery, context snapshot, budget change and termination. A crash after write intent but before its request-bound receipt is an uncertain native write and is never silently replayed. From the original manifest and seeds, reconstruction feeds recorded responses through the controller and normal tool gateway, stops before a selected decision and compares digests of physical state, actor contexts, knowledge, pending deliveries, clocks and remaining budgets. An unchanged suffix must preserve the semantic trace and outcome before the checkpoint is eligible for repair.

A repair candidate contains one or two primitives: acquire a missing native observation, redeliver acquired evidence, refresh expired evidence, route evidence to its proper owner, restore received evidence omitted from context, or request reconsideration using valid evidence. Candidates requiring hidden, future, fabricated, wrong-scope or physically late information are rejected; unknown duration remains unresolved. Redundant supersets are removed. The frozen selector orders candidates by earliest failed obligation, fewer primitives and lower native cost. After a repair, cached future responses are discarded and live execution resumes under the original remaining team budget.

4.7 Selective live verification

Treatment, irrigation, harvest, postharvest operations and an incomplete terminal finish are high impact. We compare audit-only, the existing guard, always verify, periodic verification under the same review allowance, and D-CORE witness-triggered verification. Every policy sees only the prefix. The actor's initial request is already spent; formatting correction, diagnosis, selection, native observation and reconsideration all debit the same ledger. Deferring an incomplete finish is recorded as a policy intervention rather than ordinary baseline behavior.

4.8 Algorithm and bounded properties

The offline evaluator first validates schemas and artifact bindings, reconstructs time-indexed fact eligibility and commits world branches from authoritative evidence. It then matches native events to categorical transition predicates, evaluates negative obligations and semantic paths, reconstructs local policy choices, and reports disagreement and provenance localization. It never consumes runtime transition IDs or guard judgments as truth. Audit mismatches are reported explicitly.

Three bounded properties clarify the construction. *Projection indistinguishability*: if two traces have the same projection onto actions and outcomes and different correct information-path labels, no deterministic diagnostic of that projection can recover both labels; a randomized diagnostic has the same output distribution for both. This follows directly because the input is identical. *Nonanticipation*: fixing the specification, snapshot, eligibility times, scope, deadline and channel state fixes the fact vector and hence one policy rule; future observations cannot change the earlier set of permitted responses. *Conditional serialization invariance*: fixing matching, applicability, predicates, times, weights and the dependency graph, topological reserialization of independent records preserves EF and CC. Their predicates and denominators are unchanged. This does not cover ambiguous rematching, reordered physical execution or changed graph structure. These are bounded observations with metamorphic tests, not claims of a new general theory.

5 Experimental Design

5.1 Scenarios and native sensitivity

The study retains Wet-June, Disease–Drought and Three-cultivar subject to validation. Each uses the native 64-ridge full-season environment. Wet-June tests disease reinspection, warning freshness, treatment execution windows and recovery. Disease–Drought tests diagnostic change from disease to water limitation. Three-cultivar tests region-specific treatment, irrigation and staggered harvest. Distinct outcomes are measured for biological yield and recovered marketable yield.

Before confirmation, fixed-workflow interventions check that the selected decisions have the proposed physical meaning. Wet-June includes fungicide omissions and interventions on execution windows and recovery. Three-cultivar includes fungicide, irrigation and swapped-scope interventions. Disease–Drought requires accepted irrigation, complete paired outcomes, at least 50% stressed target ridges before intervention and at least 1% omission

loss in every confirmation pair. Rainfall, root moisture, infiltration, drainage, stage and stress accumulation are inspected before revising scenario parameters. A revision receives a separate identifier; yield rewards and thresholds are not altered to force a positive result. Weak and null effects are retained. The earlier drought candidate declares a 0.4 m restricted root layer, a 38-day dry spell and a 25 mm reference pulse as modeling assumptions. A separately versioned 60 mm engineering candidate derives its dose from pre-intervention water deficits and conserves saturation overflow; it requires fresh confirmation before selection for the paper. The scripted reference may retry native harvest rejection once per day within the already authored harvest calendar ending November 2; it preserves every rejected attempt and the native 18% grain-moisture limit. This reference policy is not injected into model agents.

Native reference completion is checked through harvest and storage rather than a permissive scenario success flag alone. Bounded handling of weather and maturity must preserve every native rejection and use a declared window for observation and action. Intervention studies use a fixed workflow and therefore do not estimate how an adaptive agent would compensate for a missing action.

5.2 Treatments, worlds and compute

The whole-season assignment contains 480 and 450 primary paired seasons, 300 live-verification seasons, and 15 unreleased reserve assignments (1,245 total). The live block crosses three scenarios, ten worlds, two transports and five policies. The reserve is disabled unless a need is declared prospectively. The main ReAct backbone is fixed; a second family appears only in a prespecified focused subset. Controller-family and three/four-agent adapters are engineering compatibility checks rather than 315 primary runs. Direct and synchronous A2A baselines receive the identical serialized public task contract; an A2A label is insufficient, so observed delegation is reported separately.

Historical development uses worlds 0–9, calibration confirmation uses declared cohorts beginning with 20–24, and the final revision smoke uses newly declared worlds 70–71. The professor’s primary worlds are 100–109 and focused analyses use 100–104. Scheduler, model and fault seeds remain distinct. Deterministic named faults are not independent replicates merely because a fault-seed label changes. Weather is frozen by date before agent interaction, and paired arms must have identical exogenous-world digests.

Communication conditions compare shared state and local state, free-text and structured handoffs, and audit versus enforcement. Treatments include reliable delivery, delay within validity, delay past validity, delay past a policy deadline, drop, duplicate, reorder and a declared mixture. Phase, route and qualifying-send order define frozen targets across message representations. The injector acts only on messages agents actually send. No qualifying hand-off is a valid nonactivation outcome, distinct from a targeted fault that fails to manifest. Both remain in intention-to-treat reporting.

The pinned API model is gpt-5.4-mini-2026-03-17, with explicit endpoint and temperature in manifests. Prompt construction preserves tools, capability cards, current task, recent results and failures, accepted receipt coverage and scoped evidence. It renders

current local state once and records omitted evidence identifiers. The immutable full trace remains available for audit. Attempts to correct format or schema errors share a maximum of two retries per activation. Every underlying provider request, including planning, reflection and specialist calls, enters call and token accounting. The final implementation smoke had a prospective limit of \$25 above its recorded baseline; execution stopped when ledger review showed that earlier work had already exceeded that increment. This is separate from the professor’s full-study compute estimate.

5.3 Diagnostic validation and comparisons

Controlled positive and negative traces exercise every witness mechanism on the three authored scenarios, including multiple prerequisites, local/global disagreement, execution failure and recovery. These establish mechanical behavior; natural decisions provide the empirical diagnostic test. Before D-CORE predictions, the annotation plan freezes 120 decisions, 40 per scenario and at most two per run, using scenario, condition and execution-status strata rather than evaluator labels or yield. Shortfalls remain visible. Two reviewers label independently; a separate reviewer adjudicates after submissions are locked. Agreement is measured before adjudication.

Review packets include the proposed action, relevant local and global evidence, preceding communication and native receipts, with explicit treatment assignments and evaluator outputs withheld. Message format and native results may indirectly reveal a condition. This is partial blinding, not guaranteed concealment. Labels include failure, no failure and insufficient evidence, prerequisite readiness, supported failure categories and evidence references. A blocked proposal is an assessable proposal episode, not a physically executed action.

Every comparator receives the same versioned diagnostic packet. Path outputs include native CORE, FAIRY temporal evaluation, information-enriched CORE, MARBLE-style milestones, merged PC-KTC and average actor-local PC-KTC. Diagnosis outputs include Who&When All-at-Once, AgentRx, reviewed-constraint AgentRx, a clean DCFA-style reimplement, an independent full-information checker and D-CORE. Repair outputs include generic reconsideration, fixed rules, a DoVer adaptation and D-CORE. Methods run in isolated JSON adapters with pinned source revisions, prompts, model settings, costs and explicit unsupported outputs. Saved-trace ablations remove expiry, provenance or partial-order edges independently. All sensitivity alternatives are frozen before confirmation.

6 Statistical Analysis Plan

The world realization is the independent unit and uncertainty cluster. Decisions, repetitions and repair continuations remain nested within world. Each scenario is analyzed before pooled summaries. Primary contrasts compare paired diagnosis methods, unchanged and repaired suffixes, and the five live policies under matched worlds and transport. World pairing holds the exogenous realization fixed; stochastic model outputs need not be bitwise reproducible.

Table 1: Evaluation disagreement.

Comparison	Coverage	Disagreement
Same packet; methods pending	pending	pending

Table 2: Failure diagnosis against independent labels.

Accuracy	Agreement	Coverage	Scenario
pending	pending	pending	All; 40 decisions each

Tables report assigned runs, completed, failed and missing runs, observed fault activation, policy coverage and metric denominators before conditional scores. Incomplete harvest makes final yield unavailable. Infrastructure failures remain assigned rows without invented trace metrics. Invalid proposals, native tool failures, voluntary completion and budget termination are separate outcomes. Failed and inactive assignments are never replaced with convenient successful runs. Conditional on-activation analyses, if shown, are secondary and do not inherit the interpretation of randomized assignment because activation depends on agent behavior.

Paired differences retain all available matched units, with explicit complete-pair and missing-pair counts. Uncertainty uses bootstrap resampling of whole worlds within each scenario with 2,000 resamples, preserving repeats and within-run episodes. The primary repair outcome is continuous recovered-harvest change; useful repair is also reported at 0.5%, 1% and 2% of matched scripted reference harvest. Live verification reports the one-sided 95% bound against a -1% normalized-harvest noninferiority margin and the full two-sided interval. Paired p-values use a two-sided sign-flip test of world-level differences, exact through 16 clusters and otherwise using 2,000 draws with a plus-one correction. Holm correction applies within prespecified families. Cluster counts accompany intervals; with fewer than two clusters no interval is reported.

For human-labeled decisions, report determinate diagnosis coverage, unresolved cases, conditional and unconditional correctness, multi-label confusion matrices, and pre-adjudication agreement. Human-uncertain decisions remain visible and are excluded from determinate accuracy. Paired method comparisons use identical packets. The stratified annotation sample does not estimate population failure prevalence. Physical inputs, provider spending, setup assumptions and break-even savings are separate; labor, profitability, carbon and field-performance claims require measurements absent from this study.

7 Results: Pending

The primary experiments, repair continuations and independent annotations have not been completed. The four table shells below are generated by the analysis pipeline and contain no empirical estimates. Engineering pilots and failed attempts are archived separately and excluded.

The completed results section will report disagreement and diagnosis before matched repairs and live verification. Coverage, unresolved cases, denominators and failures accompany every

Table 3: Matched native repair continuations.

Condition	Harvest change	N
60 checkpoints; matched suffixes	pending	pending

Table 4: Selective live verification.

Cost	Coverage	Harvest	Policy
pending	pending	pending	Five policies; 300 seasons

score. Weak agreement, comparator ties, ineffective repairs, or detrimental verification require narrower conclusions. No superiority, repair-benefit or independent-validation claim is made in this draft.

8 Limitations and Reproducibility

D-CORE is conditional on an explicit specification. A complete, internally consistent specification can still encode an inappropriate agronomic assumption or exclude a legitimate adaptive response. Author definition followed by professor approval does not establish independent authorship or universal domain validity. Sensitivity analysis and independent execution annotation address different parts of this risk and cannot eliminate it.

The simulator is a reduced process model. Its synthetic intervention effects are not field treatment recommendations. Disease, drought, maturity and harvest windows can interact in ways that make a nominally important decision physically inactive. Calibration failures therefore block release rather than justify changing a reward or selecting favorable worlds. Three retained scenarios support a bounded transfer claim, not generalization across agricultural domains or arbitrary multiagent tasks.

Observation tools expose limited spatial and temporal views. A sensor average cannot establish every ridge’s state without an explicit aggregation assumption. Forecast evidence differs from contemporaneous weather. Undelivered messages are unavailable to the actor but visible to the evaluator; allowing this distinction to leak into prompts would invalidate the comparison. Free text may remain unassessable, and apparent diagnostic coverage can increase merely through richer annotation. These coverage differences must accompany accuracy comparisons.

Implementation reproducibility comprises the pinned Python version and dependency lock, exact specifications and manifests, provider endpoints, source revision, raw traces and their hashes, and deterministic analysis commands. API sampling and service behavior remain external sources of variation. Resume accepts only compatible completed runs, rejects configuration mismatch, preserves failures and does not replay uncertain native writes. A review package contains tested engineering and authored frozen artifacts with approval pending; a release package additionally requires genuine sign-off and satisfied scientific gates. Credentials and machine-specific paths are excluded from portable packages; raw artifacts are archived separately from the anonymous supplement.

AI assistance. OpenAI Codex (GPT-6, as identified by the implementation environment) assisted with code, methodological

discussion, experiment design and manuscript drafting. The supplement records the supplied design instructions and implementation plan, tool and version information and the limits of retained prompt records. Human authors must review all scientific choices, verify citations and accept responsibility for the final paper. Model-generated outputs are not independent annotations, and AI is not an author.

9 Conclusion: Empirical Claims Pending

D-CORE supplies an explicit framework for diagnosing recorded distributed information failures, selecting bounded legal repairs and testing them through matched native continuations. Its proposed study separates diagnostic validity, repair effect and live-verification cost. Final conclusions must follow the complete experiments and independent annotations, including failures, unknown cases and null or adverse effects. Acceptance readiness is not claimed by this draft.

References

- [1] Shraddha Barke, Arnav Goyal, Alind Khare, Avaljot Singh, Suman Nath, and Chetan Bansal. 2026. AgentRx: Diagnosing AI Agent Failures from Execution Trajectories. arXiv:2602.02475 <https://arxiv.org/abs/2602.02475>
- [2] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. 2025. Why Do Multi-Agent LLM Systems Fail? arXiv:2503.13657 <https://arxiv.org/abs/2503.13657>
- [3] Joseph Y. Halpern and Judea Pearl. 2005. Causes and Explanations: A Structural-Model Approach. Part I: Causes. *The British Journal for the Philosophy of Science* 56, 4 (2005), 843–887. doi:10.1093/bjps/axi147
- [4] Leslie Lamport. 1978. Time, Clocks, and the Ordering of Events in a Distributed System. *Commun. ACM* 21, 7 (1978), 558–565. doi:10.1145/359545.359563
- [5] Xixi Lu, Dirk Fahland, and Wil M. P. van der Aalst. 2015. Conformance Checking Based on Partially Ordered Event Data. In *Business Process Management Workshops*. Springer, 75–88. doi:10.1007/978-3-319-15895-2_7
- [6] Ming-Jie Ma, Jue Zhang, Fangkai Yang, Yu Kang, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. 2026. DoVer: Intervention-Driven Auto Debugging for LLM Multi-Agent Systems. In *International Conference on Learning Representations*. <https://www.microsoft.com/en-us/research/publication/dover-intervention-driven-auto-debugging-for-llm-multi-agent-systems/>
- [7] Panagiotis Michelakis, Yiannis Hadjiyiannis, and Dimitrios Stamoulis. 2025. CORE: Full-Path Evaluation of LLM Agents Beyond Final State. LAW 2025 Workshop at NeurIPS. arXiv:2509.20998 <https://arxiv.org/abs/2509.20998>
- [8] Ao Qu, Panagiotis Michelakis, Yiannis Hadjiyianni, Feng Li, Jingchi Jiang, Dimitrios Stamoulis, and Jie Liu. 2026. Full-Season Agent Evaluation in Soybean Farm Operations under Real-World Agricultural Process Dynamics. In *Second Workshop on Agents in the Wild: Safety, Security, and Beyond*. <https://openreview.net/forum?id=m49CywnvoV>
- [9] Ao Qu, Panagiotis Michelakis, Linyuan Han, Yiannis Hadjiyianni, Kun Ouyang, Konstantinos Siskos, Feng Li, Ran Meng, Jingchi Jiang, Dimitrios Stamoulis, and Jie Liu. 2026. Deploying and Evaluating a Smart-Agriculture Agentic Engine for Full-Season Soybean Farm Operations. Preprint reports acceptance to ACM SIGSPATIAL 2026. arXiv:2609.00106 <https://arxiv.org/abs/2609.00106>
- [10] Zehao Wang, Lanjun Wang, Shilong Jin, Junjie Chen, and Yanghua Xiao. 2026. DCFA: Dual-view Causal-inspired Attribution for Failure Reasoning in LLM-based Multi-agent Systems. arXiv:2609.04749 <https://arxiv.org/abs/2609.04749>
- [11] Glynn Winskel. 1989. An Introduction to Event Structures. *DAIMI Report Series* 18, 278 (1989). doi:10.7146/dpb.v18i278.6655
- [12] Shaokun Zhang, Ming Yin, Jieyu Zhang, Jiale Liu, Zhiguang Han, Jingyang Zhang, Beibin Li, Chi Wang, Huazheng Wang, Yiran Chen, and Qingyun Wu. 2025. Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems. In *Proceedings of the 42nd International Conference on Machine Learning*. arXiv:2505.00212 <https://arxiv.org/abs/2505.00212>
- [13] Kunlun Zhu, Hongyi Du, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, and Jiaxuan You. 2025. MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 8580–8622. doi:10.18653/v1/2025.acl-long.421